

Itay Yona

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Professional Experience

- 2022-Present **AI21 Labs (NLP), Algorithm Developer**
Developing State-of-the-Art Deep Learning NLP capabilities for our Large Language Models
- 2021-2022 **Israeli Ministry of Defense, Research Scientist and AI Consultant**
Designed, conducted and analyzed information from controlled laboratory-based investigations, experiments and trials on Deep Neural Networks
- 2020-2022 **Weizmann AI Center, Machine Learning Researcher**
Developed novel Causal-Discovery ML approach to datasets with multicollinearity
Combined multiple Explainable ML methods: GAMs, Bayesian statistics, Shapely values
Designed and implemented distributed system architecture (MapReduce-like)
- 2018-2019 **Pagaya (Fintech), Senior Data Scientist**
Initiated from scratch the real estate research, yielding 30% of the company revenues
- 8200 Unit, Israeli Defense Forces (NSA/GCHQ equivalent)**
- 2016–2018 **Senior Multidisciplinary Researcher - Cyber-Security & Electrical Engineering**
Solved the flagship project of the unit by bridging two seemingly unrelated disciplines
Led innovative long-term blackbox hardware research
- 2016–2016 **AHAM (Problem Solving) Research Course Instructor**
Commanded and tutored highly talented mathematicians and cyber-security specialists
- 2013–2016 **Multidisciplinary Researcher - Cyber-Security & Math**
Directed state-of-the-art mathematical and cryptographic systems research

Education

- 2020-2022 **Weizmann Institute of Science, M.Sc. Computer Science & Neuroscience (GPA: 95)**
Thesis: Analysis of High-Dimensional Neuronal Tuning Curves in Bat Hippocampus during Complex Behaviors. Advisor: Prof. Nachum Ulanovsky
Towards publication (in progress)
 - Presented on: SFN (2021), ISFN (2021), Cosyne (2022)
 - Awarded full 2-years research fellowship
- 2016-2019 **The Open University, B.Sc. Computer Science (GPA: 92)**
During the military service
 - Graduated Magna Cum Laude
 - University Presidential Award in 2018
- 2013-2013 **AHAM Intelligence Course, Research Track**
Completed prestigious exclusive 6 month trainings for (the nationally top 15) exceptional mathematicians and tech prodigies

Volunteering

2021-Present **The Problem-Solving Course, Co-Founder**

Lecturing gifted students from the Israeli periphery and tech companies
Designed course material on the principles of problem-solving
Formed partnership with IASA – a leading high school

2022-Present **Effective Altruism Israel, Coordinator of XAI reading group**

Initiated a reading group on mechanistic interpretability of Deep Neural Networks
Comprised of over 30 scholars and 7 permanent participants from a variety of disciplines:
hackers, neuroscientists and AI experts

Awards

Won 1st place in 5 major international cyber-security CTF tournaments (team Pasten):

- 32C3 (2015)
- 33C3 (2016)
- Google CTF (2016)
- 34C3 (2017)
- Google CTF (2017)

Technical Summary

Programming Skills: Python, C\C++, x86 Asm

Professional Interests: Deep Learning, Machine Learning, Algorithms, Statistics, Neuroscience, Natural Language Processing, Reinforcement Learning, Math, Software and Hardware Reverse Engineering

Frameworks: PyTorch, TensorFlow, Kubernetes